

safety valves in that a PTRV opens gradually at a set point. Normally the valve opens at 210 degrees or at 125 psi at a minimum. The pipe that carries the relieved water or steam from the tank must extend to within 18 inches of the deck to prevent a hazardous condition.

Always remember that water in a closed vessel will expand when heated and cause the pressure to rise. If, for some reason, a control fails to turn the heat source off, the pressure will be relieved by the PTRV. The water that comes out of the outlet of a PTRV will flash off due to the pressure change; steam along with scalding water will be discharged from the tank.

Test your Knowledge (Select the Correct Response)

1. What two types of urinals are most commonly used by the Seabees?
 - A. Floor- and wall-mounted
 - B. Floor- and side beam-mounted
 - C. Parallel and perpendicular mounted
 - D. Wall- and diagonal mounted

2.0.0 PLUMBING REPAIRS

This portion of the chapter will deal with some of the more common plumbing repairs that will be necessary from time to time to keep plumbing systems operating properly. Proper repairs and maintenance techniques save money by extending the life of plumbing systems. For example, one water faucet that is leaking one drop of water each second wastes about 2,300 gallons of water per year.

2.1.0 Water Breaks

Water distribution piping at one time or another will require repair on a leak or a break in the line. The following are problems you may have during a waterline break:

- The water supply for fire protection is reduced or does not exist.
- Escaping water under pressure undermines structures, damages foundations, destroys landscaping, or causes a serious erosion problem.
- A broken pipe causes a health hazard because the distribution system can become contaminated by external sources.
- The water supply for normal domestic or industrial use can be completely cut off.

To ensure proper repair of a water break, keep red line prints and as built drawings on hand that show the water distribution system, existing conditions, and locations. Ensure that you red line your set of prints every time you make a repair or line change.

Additionally, notify engineering of your line repair or line change, and they will update the master set of base prints.

At some activities, electronic devices are available for subsurface survey and pipe location work. Sometimes you may have to find points of interconnection, pipe diameters, and the condition of exterior surfaces or coatings. For future use, make notes on the maintenance prints to show the general condition of the system. Use a symbol that stands out to show the approximate age of the installation or its parts. Prints should be complete and up-to-date. In maintenance or repair, these prints help in planning maintenance. Many times they offer clues to the most probable location and

probable cause of trouble. Now and then, the system should be flushed through hydrants and blow-offs to remove scale and accumulation in pipes and fittings. When performing this operation, start at the hydrants or blow-offs nearest the source of supply to conserve water and to stir up less of the distribution system. Each point should be flushed until the water comes out reasonably clear. All valves should be in their normal operating positions before you go on to the next point. Flushing dead ends is vital. When flushing does not induce enough velocity to scour the mains clean, night flush them with a large discharge. Night operation lessens work disruption caused by water shutoff or decreased water pressure.

2.2.0 Water Mains

Since water main breaks must be repaired as fast as possible, personnel must be trained and repair plans made in advance. The following procedures are essential:

1. Post the telephone numbers of the fire department and key personnel and have alternate personnel available in case members of the regular repair crew cannot be reached at the time of a break. Notify the public works officer at the time the break is reported.
2. Always keep the following items readily available: valve keys, hand tools, digging tools, pavement breakers, trench shoring, a portable centrifugal or diaphragm pump, floodlights, an emergency chlorinator, and calcium hypochlorite powder.
3. Maintain enough pipe repair materials and supplies. As a temporary measure, wooden plugs can be used to stop small holes in a main. These plugs can be replaced later with metal plugs, or repairs may be made by other means. Wooden plugs can also be used temporarily to plug the ends of a pipe up to 8 inches in diameter, but such plugs must be braced to withstand existing main pressure. After repairs are completed, the main must be disinfected.

2.3.0 Thawing Frozen Pipes

In cold weather, a water-supply system can freeze. Because of the lack of protection against freezing, and sometimes regardless of it, pipes frequently freeze in temperate zones. When this happens, the pipes must be thawed. Breaks must be found, if possible, before natural thawing to prevent damage to material and property. Alert personnel to watch for the signs of a broken line. The prevention of freezing pipes can sometimes be accomplished by using heat tapes and cables.

Before starting to thaw a frozen pipe, open faucets affected by the freeze. Frozen pipes can be thawed by applying heat at the lowest open end of the frozen section. (Do NOT start in the middle of a frozen section because a pocket of steam could develop and an explosion or damage to the pipe can occur.) Where there is no danger of fire, simply heat the pipe with a blowtorch, applying the flame on the outside of the pipe.

Using hot water is the preferred method for thawing frozen water pipes or heating pipes inside of buildings. Do NOT use an open flame. A safe method is to wrap the frozen section of pipe with cloth and pour hot water on it until the ice gives way. Remember to protect the floor by catching the water in buckets or by covering the floor with material to absorb the water.

A good method of thawing water pipes that are underground or otherwise hard to get to is shown in *Figure 6-29*. When using this method, remove the fittings (see illustration) and insert one end of the small pipe or tube into the frozen pipe. Now add an elbow and a piece of vertical pipe to the outer end of the thaw pipe. Place a bucket under the opening to the frozen pipe and insert a funnel in the open end of the vertical pipe.

With that done, start pouring boiling water through the funnel into the pipe. As the ice melts, push the thawed pipe forward. Where necessary, add pipe at the outer end until a passage is made through the ice.

Withdraw the thaw pipe quickly after the flow starts and do not stop the flow until the thaw pipe is fully removed and the pipe cleared of ice.

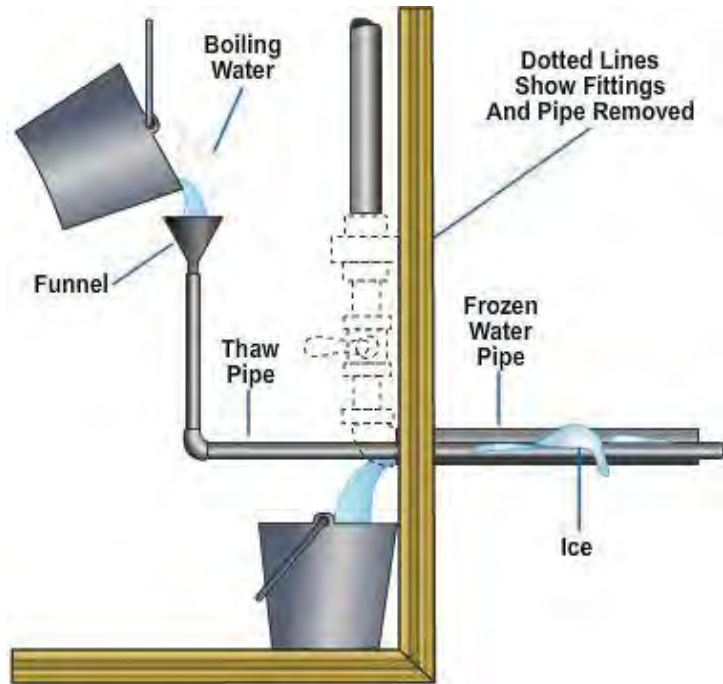


Figure 6-29 – Thawing an underground or otherwise inaccessible pipe.

Instead of a funnel, a small force pump can be used. This pump is useful for thawing a long piece of pipe. When available, you can use steam in place of hot water. The above method can also be used without the elbow and piece of vertical pipe, as shown in *Figure 6-29*. Simply connect the funnel to the outer end of the thaw pipe with rubber tubing. Have the tubing long enough so you can hold the funnel above the level of the frozen pipe. In this way, you give the hot water a head, forcing the cooled water back to the opening where it runs out into the pail. The advantage of the elbow and vertical pipe is that they increase the head of the water and make the use of the funnel easier.

2.3.1 Electrical Thawing

Electrical thawing of frozen service lines is quick and cheap. The electrical current for the thawing operation consists of a source of current (a DC generator, such as a welding outfit, or a transformer connected to an AC outlet) and two insulated wires connecting the current source and the pipe (*Figure 6-30*).

Table 6-3 – Relation of current and voltage required for thawing.

Type of pipe	Pipe Size (in.)	Pipe Length (ft.)	Approximate	Approximate (amps)
Wrought Iron	3/4	600	60	250
	1	600	60	300
	1	600	60	350
	1/2	500	55	400
	2	400	40	450
	3	400	50	500
Cast Iron	4	400	50	600
	6	300	40	600
	8			

The time for electrical thawing may vary from 5 minutes to over 2 hours, depending on pipe size and length, intensity of freezing, and other factors. The best practice is to apply current until the water flows freely.

Use the following procedures in electrical thawing:

1. **DC Generator.** To thaw pipes with a welding generator or similar DC source, set the generator to the correct amperage for the pipe to be thawed and connect the leads to the pipe.
2. **AC Circuit.** Transformers are required to adjust amperage of an AC circuit to the pipe being thawed. To reduce hazards, have a competent Construction Electrician (CE) set and connect transformers, make the connections, and assist in the thawing process. Where frequent thawing is necessary at different points, the transformers may be mounted on a trailer for ready use.

Some precautions in electrical thawing are given below:

- Avoid a higher current than listed in *Table 6-3*. When in doubt, use low current for a longer period.
- Select contact points on the pipe as close as possible to the frozen section.
- Assure that contact points are free of rust, grease, or scale.
- Remove meters, electrical ground connections, and couplings to buildings with plumbing in the pipeline to be thawed.
- If there are gaskets or other insulation at pipe joints, thaw the pipe in sections between such joints, or use copper jumpers to close the circuit across insulated points.

2.3.2 Steam Thawing

Steam thawing of frozen systems is slower than electrical thawing and should be used only when insulating materials in pipes (plastic, transite, and wood), pipe joints, or couplings make the use of electricity impractical. In steam thawing, a hose connected to a boiler is inserted through a disconnected fitting and gradually advanced as the steam melts the ice.

2.3.3 Variation of Water Pressure

A change of water pressure can cause much discomfort to persons using the plumbing system. The mixture of hot and cold water from a shower can suddenly vary in temperature or rate of flow when water is turned on at another outlet. Failure to remedy this condition could injure somebody, especially if the temperature is scalding.

When a switch in pressure and water flow occurs often, look at the water pipes. Check the pipes to see if they are the proper size in diameter for their length and height as originally installed. Also look for liming and corrosion inside the pipes. Enough liming and corrosion can reduce the diameter of the pipe, causing restrictions that lead to low pressure and slow water flow.

Sometimes the trouble occurs after more fixtures have been installed in the system. When this happens, the piping is probably overloaded because of the extra fixtures. Pressure and water flow may also change when there is too much friction in the pipe, too many fittings, or changes in the direction of the piping.

If the pressure in showers changes only when other outlets are open, you can usually correct the trouble by installing automatic mixing valves. The only answer to an increase in the water flow from pipes that are too small is to replace them with larger pipes.

Test your Knowledge (Select the Correct Response)

2. When thawing frozen water pipes or heating pipes inside buildings, which method is preferred?
- A. Open flame
 - B. Hot Water
 - C. Electrical thawing unit
 - D. Steam thawing unit

3.0.0 PIPE LEAKS

When a leak develops at a threaded joint of pipe, one of the most likely suspects is a fractured or ruptured pipe. Fractures often occur at the end of a length of pipe because of strain imposed by vibration of water hammer. It occurs at the end of the pipe because the wall thickness is decreased and weakened by threading. The risk of fracture becomes even greater when the threads are not cut true. In cold climates, freezing sometimes causes pipes to rupture, in which case replacement becomes necessary. A loose or cracked fitting can also cause leakage at the threaded joint of a pipe. These and other common failures resulting in pipe leakage make it important for you to determine the exact location and cause of failure before beginning any repairs to the piping.

3.1.0 Locating Leaks

Find and repair leaks in the water piping system as quickly as possible to prevent serious damage to footings, walls, floors, plaster, and other parts of the structure, and to conserve water. Also, sanitation and hygiene issues are associated with water leaks, such as mold, insect's sanitary system and disease. Find leaks systematically by inspecting exposed piping and valves and by examining walls, floors, and ceilings around concealed piping. You should also check gauges, meters, and other water flow recording devices for evidence of abnormal flow, which might reveal loss through leakage.